

PLANAR POINT SETS WITH SMALL DISTINCT-DISTANCE SUBSETS

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ABSTRACT. Let $F_2(n)$ denote the minimum, over all n -point sets in the plane, of the largest size of a subset whose pairwise distances are all distinct. The classical square-grid construction gives $F_2(n) \ll n^{1/2}/(\log n)^{1/4}$. We prove that this can be improved by a fixed power: there is an absolute constant $\varepsilon > 0$ such that $F_2(n) \ll n^{1/2-\varepsilon}$. The proof combines two ingredients: the number-field towers of bounded root discriminant used in the OpenAI unit-distance construction, and an argument of Pohoata and Sheffer based on split primes.

1. INTRODUCTION

For $n \geq 1$, let $F_2(n)$ be the minimum, over all n -point sets $P \subset \mathbb{R}^2$, of the maximum size of a subset of P whose pairwise distances are all distinct. We call such a subset *distance-Sidon*. Equivalently, every n -point set in the plane contains a distance-Sidon subset of size at least $F_2(n)$. The problem is to determine the order of magnitude of $F_2(n)$ as $n \rightarrow \infty$. This is the planar case of Erdős' distance-Sidon subset problem [2], rooted in the classical study of distances determined by finite point sets [7], [8]; see also [11] and [5].

Until recently, the long-standing bounds were

$$\frac{n^{1/3}}{(\log n)^{1/3}} \ll F_2(n) \ll \frac{n^{1/2}}{(\log n)^{1/4}}.$$

The upper bound comes from the square integer grid: a $\sqrt{n} \times \sqrt{n}$ grid determines only $O(n/\sqrt{\log n})$ squared distances, by the classical estimate for integers representable as a sum of two squares, and hence no distance-Sidon subset of the grid can have more than $O(n^{1/2}/(\log n)^{1/4})$ points. The lower bound was proved by Charalambides [3], using the Guth–Katz distinct-distance machinery [9]. Recently Clemen, Führer and Roche-Newton [4] removed the logarithmic loss on the lower-bound side, proving $F_2(n) \gg n^{1/3}$.

Our main result is a polynomial improvement over the upper bound.

Theorem 1.1. *There exists an absolute constant $\varepsilon > 0$ such that*

$$F_2(n) \ll n^{1/2-\varepsilon}.$$

The proof combines two recent ideas. The first is the number-field framework used in the OpenAI construction disproving Erdős' unit-distance conjecture [14], [1]. It provides totally real fields L of growing degree and bounded root discriminant, together with rational primes $q \equiv 1 \pmod{4}$ which split completely in L . We

use these fields and split primes to construct point sets with small distance-Sidon subsets.

The second idea is the split-prime isotropic-line sieve of Pohoata and Sheffer for the integer grid [16]. In their argument one chooses primes $p_i \equiv 1 \pmod{4}$, roots $x_i^2 \equiv -1 \pmod{p_i}$, and for each sign vector $\sigma \in \{\pm 1\}^t$ considers the congruence system

$$u \equiv \sigma_i x_i v \pmod{p_i} \quad (1 \leq i \leq t).$$

These are isotropic directions modulo $d = \prod_i p_i$, and a double count over the 2^t sign vectors bounds distance-Sidon subsets of $[N]^2$ by

$$N \exp\left(-c \frac{\log N}{\log \log N}\right)$$

for an absolute constant $c > 0$. See also the combinatorial large-sieve framework announced in [6] and the related split-prime amplification in [15].

Our argument lifts this sieve from $\mathbb{Z}^2$ to $\mathcal{O}_L^2$. If q_i splits completely in L , then it gives $[L : \mathbb{Q}]$ prime ideals, and therefore the 2^t local sign choices in the grid become $2^{t[L:\mathbb{Q}]}$ choices. This tower amplification is the main point: a fixed set of split primes yields a saving exponential in the field degree, and that exponential-in-degree saving becomes a fixed power saving in the number of planar points.

Organization. Section 2 records the required facts about the number-field tower. Section 3 constructs the planar point sets from algebraic boxes. Section 4 proves the tower-amplified split-prime sieve. Section 5 chooses the parameters and proves Theorem 1.1.

Notation. For a finite set S , we write $\#S$ for its cardinality.

2. PRELIMINARIES

2.1. Basic algebraic number theory. We use standard terminology from algebraic number theory; see, for example, [13, Chapter I] or [12].

A *number field* F is a finite extension of $\mathbb{Q}$. Put $d = [F : \mathbb{Q}]$, and let $\sigma_1, \dots, \sigma_d : F \hookrightarrow \mathbb{C}$ be the embeddings of F into $\mathbb{C}$. For $\alpha \in F$, the *trace* of α is

$$\mathrm{Tr}_{F/\mathbb{Q}}(\alpha) = \sum_{k=1}^d \sigma_k(\alpha).$$

Let $\mathcal{O}_F$ be the ring of integers of F , i.e.,

$$\mathcal{O}_F = \{x \in F : x \text{ is a zero of some monic polynomial over } \mathbb{Z}\}.$$

Then $\mathcal{O}_F$ has a $\mathbb{Z}$ -basis $\omega_1, \dots, \omega_d$. The *discriminant* of F is

$$\Delta_F = \det(\mathrm{Tr}_{F/\mathbb{Q}}(\omega_i \omega_j))_{1 \leq i, j \leq d}.$$

This integer is independent of the choice of integral basis. The *root discriminant* of F is

$$|\Delta_F|^{1/[F:\mathbb{Q}]}.$$

A number field F is said to be *totally real* if every embedding $F \hookrightarrow \mathbb{C}$ has image in $\mathbb{R}$.

A *rational prime* is an ordinary prime number in $\mathbb{Z}$. Let $F/\mathbb{Q}$ be a number field of degree d , and let q be a rational prime. The ring $\mathcal{O}_F$ is a Dedekind domain, so every nonzero ideal factors uniquely as a product of prime ideals, and every

nonzero prime ideal is maximal; see, for example, [12, Theorem 3.29, Theorem 3.7, and Definition 3.3]. We say that q *splits completely* in F if

$$q\mathcal{O}_F = \mathfrak{q}_1 \cdots \mathfrak{q}_d$$

as a product of d distinct prime ideals of $\mathcal{O}_F$. In this case $\mathcal{O}_F/\mathfrak{q}_k \cong \mathbb{F}_q$ for $k = 1, \dots, d$.¹ For an infinite algebraic extension $M/\mathbb{Q}$, the prime q is said to split completely in M if it splits completely in every finite subextension of M .

2.2. Pro- p extensions. We use standard terminology for pro- p groups; see, for example, [17, Chapters 1, 2, and 4].

An inverse system consists of groups G_i and homomorphisms $\phi_{ij} : G_j \rightarrow G_i$ for $i \leq j$, satisfying $\phi_{ii} = \text{id}$ and $\phi_{ik} = \phi_{ij} \circ \phi_{jk}$ for $i \leq j \leq k$. Its inverse limit is the group of tuples $(g_i)_i \in \prod_i G_i$ such that $\phi_{ij}(g_j) = g_i$ for all $i \leq j$. A *pro- p group* is an inverse limit of finite p -groups, where each finite p -group is given the discrete topology and the inverse limit is given the subspace topology inherited from $\prod_i G_i$. A *Galois pro- p extension* M/K is a Galois extension whose Galois group is a pro- p group.

2.3. A tower of totally real fields. [1, Proposition 2.3] states the following result.

Proposition 2.1. *There exists an infinite tower of totally real fields over $\mathbb{Q}$ with bounded root discriminant and with infinitely many completely split primes $q \equiv 1 \pmod{4}$.*

Although the following proposition is not stated separately in [1], it follows because the proof of [1, Proposition 2.3] obtains the tower from an infinite Galois pro-2 extension of $\mathbb{Q}$, constructed using [10, Theorem 2.8 of the arXiv version, or Theorem 4 of the published version].

Proposition 2.2. *There exists an infinite tower $\mathbb{Q} = F_0 \subset F_1 \subset F_2 \subset \cdots$ of totally real fields over $\mathbb{Q}$ such that the following hold.*

- (1) *The tower has bounded root discriminant.*
- (2) *Infinitely many rational primes $q \equiv 1 \pmod{4}$ split completely in every F_j .*
- (3) *$[F_j : \mathbb{Q}] = 2^j$ for every $j \geq 0$.*

Proof. The proof of Proposition 2.1 given in [1] supplies an infinite Galois pro-2 extension $M/\mathbb{Q}$ whose intermediate fields E with $\mathbb{Q} \subset E \subset M$ and $[E : \mathbb{Q}] < \infty$ form a tower of totally real fields with bounded root discriminant and with infinitely many completely split primes $q \equiv 1 \pmod{4}$.

It remains only to choose finite layers with degree exactly 2^j . Let $G = \text{Gal}(M/\mathbb{Q})$. Set $G_0 = G$, and suppose G_j has been chosen. Since G_j is an infinite pro-2 group, it has an open subgroup $G_{j+1} \subset G_j$ of index 2: take a nontrivial finite 2-group quotient of G_j , then take the inverse image of an index-2 subgroup. Put $F_j = M^{G_j}$.² Then

$$[F_{j+1} : F_j] = [G_j : G_{j+1}] = 2,$$

so $[F_j : \mathbb{Q}] = 2^j$. □

¹Indeed, if $q\mathcal{O}_F = \prod_k \mathfrak{q}_k^{e_k}$ and $\#(\mathcal{O}_F/\mathfrak{q}_k) = q^{f_k}$, then $\#(\mathcal{O}_F/q\mathcal{O}_F) = q^d$ and the Chinese remainder theorem gives $d = \sum_k e_k f_k$. In the completely split case there are d distinct prime ideals and each $e_k = 1$, so every $f_k = 1$.

²Here $M^{G_j} = \{x \in M : \sigma(x) = x \text{ for every } \sigma \in G_j\}$.

3. THE POINT SET CONSTRUCTION

We fix the tower $\mathbb{Q} = F_0 \subset F_1 \subset F_2 \subset \cdots$ from Proposition 2.2. Choose $D \geq 1$ such that every field in this tower has root discriminant at most D .

3.1. A translated algebraic box. Let $j \geq 0$. Write the real embeddings of F_j as

$$(1) \quad \sigma_{j,1}, \dots, \sigma_{j,2^j} : F_j \hookrightarrow \mathbb{R}$$

and let

$$(2) \quad \Sigma_j : F_j \rightarrow \mathbb{R}^{2^j}, \quad \Sigma_j(\alpha) = (\sigma_{j,1}(\alpha), \dots, \sigma_{j,2^j}(\alpha)).$$

Then $\Sigma_j(\mathcal{O}_{F_j})$ is a lattice of covolume³

$$(3) \quad \text{covol}(\Sigma_j(\mathcal{O}_{F_j})) = |\Delta_{F_j}|^{1/2} \leq D^{2^j/2}.$$

Fix $R \geq 1$. Average $\#(\Sigma_j(\mathcal{O}_{F_j}) \cap (\tau + [-R/2, R/2]^{2^j}))$ as τ ranges over a fundamental domain for $\Sigma_j(\mathcal{O}_{F_j})$. By (3), the average is

$$\frac{\text{vol}([-R/2, R/2]^{2^j})}{\text{covol}(\Sigma_j(\mathcal{O}_{F_j}))} = \frac{R^{2^j}}{|\Delta_{F_j}|^{1/2}} \geq \left(\frac{R}{\sqrt{D}}\right)^{2^j}.$$

Thus there exists $\tau_{j,R} \in \mathbb{R}^{2^j}$ such that

$$\#(\Sigma_j(\mathcal{O}_{F_j}) \cap (\tau_{j,R} + [-R/2, R/2]^{2^j})) \geq \left(\frac{R}{\sqrt{D}}\right)^{2^j}.$$

Define

$$(4) \quad B_{j,R} := \{a \in \mathcal{O}_{F_j} : \Sigma_j(a) \in \tau_{j,R} + [-R/2, R/2]^{2^j}\}.$$

Then

$$(5) \quad \#B_{j,R} \geq \left(\frac{R}{\sqrt{D}}\right)^{2^j}.$$

For every $a, c \in B_{j,R}$, the points $\Sigma_j(a)$ and $\Sigma_j(c)$ lie in the same translate of $[-R/2, R/2]^{2^j}$, and hence

$$(6) \quad |\sigma_{j,k}(a - c)| \leq R, \quad k = 1, \dots, 2^j.$$

3.2. Projection to the plane. For j and R , define

$$(7) \quad P_{j,R} = \{(\sigma_{j,1}(a), \sigma_{j,1}(b)) : a, b \in B_{j,R}\} \subset \mathbb{R}^2.$$

Since $\sigma_{j,1}$ is injective, (5) gives

$$(8) \quad \#P_{j,R} = (\#B_{j,R})^2 \geq \left(\frac{R}{\sqrt{D}}\right)^{2^{j+1}}.$$

Note that the map $B_{j,R}^2 \rightarrow P_{j,R}$, $(a, b) \mapsto (\sigma_{j,1}(a), \sigma_{j,1}(b))$, is a bijection. Thus a subset $S \subset P_{j,R}$ corresponds to its inverse image

$$(9) \quad \{(a, b) \in B_{j,R}^2 : (\sigma_{j,1}(a), \sigma_{j,1}(b)) \in S\} \subset B_{j,R}^2.$$

³Covolume is the volume of a fundamental domain. Since F_j is totally real, if $\omega_1, \dots, \omega_{2^j}$ is a $\mathbb{Z}$ -basis of $\mathcal{O}_{F_j}$ and $A = (\sigma_{j,k}(\omega_\ell))_{1 \leq k, \ell \leq 2^j}$, then A is the real coordinate matrix of the lattice basis $\Sigma_j(\omega_1), \dots, \Sigma_j(\omega_{2^j})$, so $\text{covol}(\Sigma_j(\mathcal{O}_{F_j})) = |\det A|$. Also $(\text{Tr}_{F_j/\mathbb{Q}}(\omega_k \omega_\ell))_{1 \leq k, \ell \leq 2^j} = A^T A$. Hence $|\Delta_{F_j}| = |\det(A^T A)| = |\det A|^2$.

In the proof of Theorem 1.1, we choose j and R suitably and use $P_{j,R}$ as the ambient planar set.

Proposition 3.1. *Let $S \subset P_{j,R}$, and suppose that all pairwise distances in S are distinct. Let A be the inverse image of S , as in (9). Then, for every nonzero $\eta \in \mathcal{O}_{F_j}$,*

$$(10) \quad \#\{((a,b), (c,d)) \in A \times A : (a-c)^2 + (b-d)^2 = \eta\} \leq 2.$$

Proof. Let (a,b) and (c,d) be elements of A . The squared Euclidean distance between their projections is

$$(11) \quad (\sigma_{j,1}(a) - \sigma_{j,1}(c))^2 + (\sigma_{j,1}(b) - \sigma_{j,1}(d))^2 = \sigma_{j,1}((a-c)^2 + (b-d)^2).$$

If $(a-c)^2 + (b-d)^2 = \eta$, then this squared distance is $\sigma_{j,1}(\eta)$. Since $\eta \neq 0$, the two projected endpoints are distinct. Since all pairwise distances in S are distinct, there is at most one unordered pair at this distance, hence at most two ordered pairs. $\square$

4. THE SIEVE ESTIMATE

We bound the size of a subset of $P_{j,R}$ whose pairwise distances are all distinct.

4.1. Counting elements in an ideal. For a nonzero ideal $\mathfrak{a} \subset \mathcal{O}_{F_j}$, write $N\mathfrak{a} = \#(\mathcal{O}_{F_j}/\mathfrak{a})$.⁴ For $\alpha \in F_j$, using the embeddings $\sigma_{j,k} : F_j \hookrightarrow \mathbb{R}$ from (1), write $N_{F_j/\mathbb{Q}}(\alpha) = \prod_{k=1}^{2^j} \sigma_{j,k}(\alpha)$.⁵ Recall from (2) that $\Sigma_j(\alpha) = (\sigma_{j,1}(\alpha), \dots, \sigma_{j,2^j}(\alpha))$.

Lemma 4.1. *Fix $j \geq 0$, a nonzero ideal $\mathfrak{a} \subset \mathcal{O}_{F_j}$, and $Y > 0$. Then*

$$\#\{\eta \in \mathfrak{a} : \Sigma_j(\eta) \in [-Y, Y]^{2^j}\} \leq \left(1 + \frac{2Y}{(N\mathfrak{a})^{1/2^j}}\right)^{2^j}.$$

Proof. If $\eta_1 \neq \eta_2$ are two elements of $\mathfrak{a}$, then $\eta_1 - \eta_2$ is a nonzero element of $\mathfrak{a}$. Since $(\eta_1 - \eta_2) \in \mathfrak{a}$, we have

$$|N_{F_j/\mathbb{Q}}(\eta_1 - \eta_2)| = N((\eta_1 - \eta_2)) \geq N\mathfrak{a}.$$

On the other hand,

$$|N_{F_j/\mathbb{Q}}(\eta_1 - \eta_2)| = \prod_{k=1}^{2^j} |\sigma_{j,k}(\eta_1 - \eta_2)|.$$

Since this product has 2^j factors and is at least $N\mathfrak{a}$, for some k we have

$$|\sigma_{j,k}(\eta_1) - \sigma_{j,k}(\eta_2)| = |\sigma_{j,k}(\eta_1 - \eta_2)| \geq (N\mathfrak{a})^{1/2^j}.$$

Therefore the points $\Sigma_j(\eta)$, with $\eta \in \mathfrak{a}$, are separated in the sup norm by $(N\mathfrak{a})^{1/2^j}$. This proves the lemma. $\square$

⁴See, e.g., [12, Section 4, “Norms of ideals,” before Proposition 4.2].

⁵See, e.g., [12, Section 2, “Review of norms and traces,” especially Corollary 2.20].

4.2. The sieve argument. Let $t \geq 1$ be an integer. By Proposition 2.2, we can choose distinct rational primes $q_1, \dots, q_t$ such that each $q_i \equiv 1 \pmod{4}$ and each q_i splits completely in every F_j . Put

$$Q = q_1 \cdots q_t.$$

Since q_i splits completely in F_j ,

$$q_i \mathcal{O}_{F_j} = \mathfrak{q}_{i,j,1} \cdots \mathfrak{q}_{i,j,2^j},$$

where $\mathcal{O}_{F_j}/\mathfrak{q}_{i,j,k} \cong \mathbb{F}_{q_i}$ for $k = 1, \dots, 2^j$. (See Section 2.1.) Since $q_i \equiv 1 \pmod{4}$, choose $x_i \in \mathbb{F}_{q_i}$ with $x_i^2 = -1$. For each k , the inclusion $\mathbb{Z} \subset \mathcal{O}_{F_j}$ induces an isomorphism $\mathbb{F}_{q_i} = \mathbb{Z}/q_i\mathbb{Z} \rightarrow \mathcal{O}_{F_j}/\mathfrak{q}_{i,j,k}$, and we use x_i for the image of x_i under this isomorphism.

For each sign vector

$$\theta = (\theta_{i,k})_{1 \leq i \leq t, 1 \leq k \leq 2^j} \in \{\pm 1\}^{t \cdot 2^j},$$

define the additive subgroup

$$(12) \quad L_{\theta,j} = \{(u, v) \in \mathcal{O}_{F_j}^2 : u \equiv \theta_{i,k} x_i v \pmod{\mathfrak{q}_{i,j,k}} \text{ for every } i, k\}.$$

Lemma 4.2. *For every sign vector θ , we have $[\mathcal{O}_{F_j}^2 : L_{\theta,j}] = Q^{2^j}$. Moreover, if $(u, v) \in L_{\theta,j}$, then $Q \mathcal{O}_{F_j} \mid u^2 + v^2$.*

Proof. For each prime ideal $\mathfrak{q}_{i,j,k}$, the congruence

$$u \equiv \theta_{i,k} x_i v \pmod{\mathfrak{q}_{i,j,k}}$$

is the kernel of a surjective additive homomorphism

$$(\mathcal{O}_{F_j}/\mathfrak{q}_{i,j,k})^2 \rightarrow \mathcal{O}_{F_j}/\mathfrak{q}_{i,j,k}, \quad (u, v) \mapsto u - \theta_{i,k} x_i v.$$

It therefore has index q_i modulo $\mathfrak{q}_{i,j,k}$. The Chinese remainder theorem over all pairs (i, k) gives

$$[\mathcal{O}_{F_j}^2 : L_{\theta,j}] = \prod_{i=1}^t \prod_{k=1}^{2^j} q_i = Q^{2^j}.$$

If $(u, v) \in L_{\theta,j}$, then for every i, k ,

$$u^2 + v^2 \equiv (\theta_{i,k}^2 x_i^2 + 1)v^2 = 0 \pmod{\mathfrak{q}_{i,j,k}}.$$

Since $Q = q_1 \cdots q_t$, we have

$$Q \mathcal{O}_{F_j} = \prod_{i=1}^t \prod_{k=1}^{2^j} \mathfrak{q}_{i,j,k}.$$

Since the prime ideals $\mathfrak{q}_{i,j,k}$ are distinct, their intersection is their product. Hence $Q \mathcal{O}_{F_j}$ divides $u^2 + v^2$. $\square$

Proposition 4.3. *For t sufficiently large, there exists a constant $0 < \rho < 1$ such that the following holds. If $R \geq Q$ and $S \subset P_{j,R}$ has all pairwise distances distinct, then*

$$(13) \quad \#S \leq Q^{2^j} + \left(\frac{R}{\sqrt{D}} \right)^{2^j} \rho^{2^j}.$$

Proof. Let $A \subset B_{j,R}^2$ be the inverse image of S under the bijection $B_{j,R}^2 \rightarrow P_{j,R}$ from (9). Put $\Lambda = \prod_{i=1}^t (1 + 1/q_i)$. Choose t large enough so that

$$(14) \quad \rho := \sqrt{D} \left(\frac{10\Lambda}{2^t} \right)^{1/2} < 1.$$

This is possible because every $q_i \geq 5$, so $\Lambda \leq (6/5)^t$, and therefore $10D\Lambda/2^t \leq 10D(3/5)^t \rightarrow 0$. By (14),

$$R^{2^j} \left(\frac{10\Lambda}{2^t} \right)^{2^j/2} = \left(\frac{R}{\sqrt{D}} \right)^{2^j} \rho^{2^j}.$$

Thus, since $\#S = \#A$, (13) follows from

$$(15) \quad \#A \leq Q^{2^j} + R^{2^j} \left(\frac{10\Lambda}{2^t} \right)^{2^j/2}.$$

It remains to prove (15). If $\#A < Q^{2^j}$, then (15) is immediate. Thus assume from now on that $\#A \geq Q^{2^j}$.

By Lemma 4.2, $[\mathcal{O}_{F_j}^2 : L_{\theta,j}] = Q^{2^j}$. If $\#A$ is much larger than Q^{2^j} , then there must be many $(p, p') \in A \times A$ with $p \neq p'$ and $p - p' \in L_{\theta,j}$. Accordingly, we count triples

$$(16) \quad T = \#\{(p, p', \theta) \in A \times A \times \{\pm 1\}^{t2^j} : p \neq p', p - p' \in L_{\theta,j}\}.$$

For a fixed sign vector θ , let $C_1, \dots, C_{Q^{2^j}}$ be the $L_{\theta,j}$ -cosets in $\mathcal{O}_{F_j}^2$. Then

$$A = \bigsqcup_{s=1}^{Q^{2^j}} (A \cap C_s).$$

Put $n_s = \#(A \cap C_s)$. Then $\sum_{s=1}^{Q^{2^j}} n_s = \#A$, and the number of $(p, p') \in A \times A$ with $p \neq p'$ and $p - p' \in L_{\theta,j}$ is

$$\sum_{s=1}^{Q^{2^j}} n_s(n_s - 1) = \sum_{s=1}^{Q^{2^j}} n_s^2 - \#A \geq \frac{(\#A)^2}{Q^{2^j}} - \#A = \frac{\#A(\#A - Q^{2^j})}{Q^{2^j}}.$$

Summing over the 2^{t2^j} sign vectors gives

$$(17) \quad T \geq 2^{t2^j} \frac{\#A(\#A - Q^{2^j})}{Q^{2^j}}.$$

We next prove an upper bound for T . For $(u, v) \in \mathcal{O}_{F_j}^2$, set

$$r(u, v) = \#\{(p, p') \in A \times A : p - p' = (u, v)\},$$

$$w(u, v) = \#\{\theta \in \{\pm 1\}^{t2^j} : (u, v) \in L_{\theta,j}\}.$$

By (16),

$$(18) \quad T = \sum_{(u,v) \neq (0,0)} r(u, v)w(u, v).$$

By (12),

$$w(u, v) = \prod_{i=1}^t \prod_{k=1}^{2^j} w_{i,j,k}(u, v),$$

where

$$w_{i,j,k}(u, v) = \#\{\theta \in \{\pm 1\} : u \equiv \theta x_i v \pmod{\mathfrak{q}_{i,j,k}}\}.$$

Suppose $w(u, v) > 0$. Let

$$E = \{(i, k) : 1 \leq i \leq t, 1 \leq k \leq 2^j, w_{i,j,k}(u, v) = 2\}.$$

Then $w(u, v) = 2^{\#E}$. Since $w(u, v) > 0$, every $\mathfrak{q}_{i,j,k}$ has $w_{i,j,k}(u, v) \geq 1$, and hence $\mathfrak{q}_{i,j,k} \mid u^2 + v^2$. If $(i, k) \in E$, then

$$u \equiv x_i v \pmod{\mathfrak{q}_{i,j,k}}, \quad u \equiv -x_i v \pmod{\mathfrak{q}_{i,j,k}}.$$

Subtracting gives $2x_i v \equiv 0 \pmod{\mathfrak{q}_{i,j,k}}$. Since $2x_i \neq 0$ in $\mathcal{O}_{F_j}/\mathfrak{q}_{i,j,k} \cong \mathbb{F}_{q_i}$, this gives $v \equiv 0 \pmod{\mathfrak{q}_{i,j,k}}$, and then $u \equiv 0 \pmod{\mathfrak{q}_{i,j,k}}$. Thus $\mathfrak{q}_{i,j,k}^2 \mid u^2 + v^2$ for every $(i, k) \in E$. For every subset $E' \subset E$, put

$$\mathfrak{b} = \prod_{(i,k) \in E'} \mathfrak{q}_{i,j,k}.$$

Then $\mathfrak{b} \mid Q\mathcal{O}_{F_j}$ and $Q\mathfrak{b} \mid u^2 + v^2$. Therefore we obtain

$$(19) \quad w(u, v) \leq 2^{\#E} \leq \#\{\mathfrak{b} \mid Q\mathcal{O}_{F_j} : Q\mathfrak{b} \mid u^2 + v^2\}.$$

The bound (19) is immediate if $w(u, v) = 0$.

By Proposition 3.1, for every nonzero $\eta \in \mathcal{O}_{F_j}$,

$$(20) \quad \sum_{u^2+v^2=\eta} r(u, v) \leq 2.$$

Also, since F_j is totally real, $u^2 + v^2 = 0$ implies $u = v = 0$; hence $\eta = 0$ contributes nothing to T , where $(u, v) \neq (0, 0)$.

For $(u, v) \in A - A$, (6) gives

$$|\sigma_{j,k}(u)| \leq R, \quad |\sigma_{j,k}(v)| \leq R, \quad k = 1, \dots, 2^j.$$

Therefore

$$(21) \quad \Sigma_j(u^2 + v^2) \in [-2R^2, 2R^2]^{2^j}.$$

Combining (18), (19), (20), and (21) gives

$$\begin{aligned} T &= \sum_{(u,v) \neq (0,0)} r(u, v) w(u, v) \\ &\leq \sum_{(u,v) \neq (0,0)} r(u, v) \sum_{\mathfrak{b} \mid Q\mathcal{O}_{F_j}, Q\mathfrak{b} \mid u^2+v^2} 1 \\ &\leq \sum_{\mathfrak{b} \mid Q\mathcal{O}_{F_j}} \sum_{\substack{0 \neq \eta \in Q\mathfrak{b} \\ \Sigma_j(\eta) \in [-2R^2, 2R^2]^{2^j}}} \sum_{u^2+v^2=\eta} r(u, v) \\ (22) \quad &\leq 2 \sum_{\mathfrak{b} \mid Q\mathcal{O}_{F_j}} \#\{\eta \in Q\mathfrak{b} : \Sigma_j(\eta) \in [-2R^2, 2R^2]^{2^j}\}. \end{aligned}$$

By multiplicativity of the ideal norm, the factorization

$$Q\mathcal{O}_{F_j} = \prod_{i=1}^t \prod_{k=1}^{2^j} \mathfrak{q}_{i,j,k}$$

and the equality $N\mathbf{q}_{i,j,k} = \#(\mathcal{O}_{F_j}/\mathbf{q}_{i,j,k}) = \#\mathbb{F}_{q_i} = q_i$ give

$$N(Q\mathcal{O}_{F_j}) = \prod_{i=1}^t \prod_{k=1}^{2^j} q_i = Q^{2^j}.$$

Thus

$$(23) \quad (N(Q\mathbf{b}))^{1/2^j} = (N(Q\mathcal{O}_{F_j})N\mathbf{b})^{1/2^j} = Q(N\mathbf{b})^{1/2^j}.$$

Also, since $\mathbf{b} \mid Q\mathcal{O}_{F_j}$, we have $(N\mathbf{b})^{1/2^j} \leq Q$. Because $R \geq Q$,

$$(24) \quad \frac{R^2}{Q(N\mathbf{b})^{1/2^j}} \geq 1.$$

Applying Lemma 4.1 to $\mathbf{a} = Q\mathbf{b}$ and $Y = 2R^2$, and using (23) and (24), we get

$$\#\{\eta \in Q\mathbf{b} : \Sigma_j(\eta) \in [-2R^2, 2R^2]^{2^j}\} \leq \left(1 + \frac{4R^2}{(N(Q\mathbf{b}))^{1/2^j}}\right)^{2^j} \leq \left(\frac{5R^2}{Q}\right)^{2^j} \frac{1}{N\mathbf{b}}.$$

Substituting this into (22) gives

$$T \leq \left(\frac{10R^2}{Q}\right)^{2^j} \sum_{\mathbf{b} \mid Q\mathcal{O}_{F_j}} \frac{1}{N\mathbf{b}}.$$

Because $N\mathbf{q}_{i,j,k} = q_i$ for every k ,

$$\sum_{\mathbf{b} \mid Q\mathcal{O}_{F_j}} \frac{1}{N\mathbf{b}} = \prod_{i=1}^t \prod_{k=1}^{2^j} \left(1 + \frac{1}{N\mathbf{q}_{i,j,k}}\right) = \prod_{i=1}^t \prod_{k=1}^{2^j} \left(1 + \frac{1}{q_i}\right) = \Lambda^{2^j}.$$

Hence

$$(25) \quad T \leq \left(\frac{10R^2\Lambda}{Q}\right)^{2^j}.$$

Comparing (17) and (25), we obtain

$$2^{t2^j} \frac{\#A(\#A - Q^{2^j})}{Q^{2^j}} \leq \left(\frac{10R^2\Lambda}{Q}\right)^{2^j}.$$

Therefore

$$\#A(\#A - Q^{2^j}) \leq \left(\frac{10R^2\Lambda}{2^t}\right)^{2^j}.$$

Since $\#A \geq Q^{2^j}$, we have $(\#A - Q^{2^j})^2 \leq \#A(\#A - Q^{2^j})$, and hence

$$\#A - Q^{2^j} \leq R^{2^j} \left(\frac{10\Lambda}{2^t}\right)^{2^j/2}.$$

This proves (15). □

5. PROOF OF THEOREM 1.1

Choose t large enough so that Proposition 4.3 applies. We use the corresponding primes $q_1, \dots, q_t$, the product $Q = q_1 \cdots q_t$, and the constant $0 < \rho < 1$ from Section 4.2. Choose a real number $a > \log Q$.

Let n be sufficiently large. Let $j \geq 0$ be the integer satisfying

$$(26) \quad \frac{\log n}{4a} < 2^j \leq \frac{\log n}{2a}.$$

Put $R = \sqrt{D} n^{1/2^{j+1}}$. The upper bound in (26) gives

$$\log \left(\frac{R}{\sqrt{D}} \right) = \frac{\log n}{2^{j+1}} \geq a,$$

so $R \geq \sqrt{D} e^a \geq Q$.

Let $B_{j,R}$ and $P_{j,R}$ be as in (4) and (7). By (8),

$$\#P_{j,R} \geq \left(\frac{R}{\sqrt{D}} \right)^{2^{j+1}} = n.$$

Let $P_n \subset P_{j,R}$ be any n -point subset.

Let $S \subset P_n$ have all pairwise distances distinct. Since $S \subset P_{j,R}$, Proposition 4.3 gives

$$\#S \leq Q^{2^j} + \left(\frac{R}{\sqrt{D}} \right)^{2^j} \rho^{2^j} = Q^{2^j} + n^{1/2} \rho^{2^j}.$$

By (26),

$$Q^{2^j} = \exp(2^j \log Q) \leq \exp \left(\frac{\log Q}{2a} \log n \right) = n^{\log Q/(2a)}$$

and

$$n^{1/2} \rho^{2^j} = n^{1/2} \exp(2^j \log \rho) \leq n^{1/2} \exp \left(\frac{\log \rho}{4a} \log n \right) = n^{1/2 + \log \rho/(4a)},$$

we may choose $\varepsilon > 0$ such that

$$\frac{\log Q}{2a} \leq \frac{1}{2} - \varepsilon, \quad \frac{1}{2} + \frac{\log \rho}{4a} \leq \frac{1}{2} - \varepsilon.$$

Then

$$\#S \leq 2n^{1/2-\varepsilon}.$$

Thus every subset of P_n with all pairwise distances distinct has size at most $2n^{1/2-\varepsilon}$. Consequently

$$F_2(n) \leq 2n^{1/2-\varepsilon}$$

for all sufficiently large n .

USE OF AI

During the preparation of this work, the author used OpenAI's GPT-5.5 Pro to explore possible proof strategies and to obtain methodological ideas behind the argument. After using this tool, the author independently verified the mathematical details, refined the exposition, and takes full responsibility for the final manuscript.

REFERENCES

1. Noga Alon, Thomas F. Bloom, W. T. Gowers, Daniel Litt, Will Sawin, Arul Shankar, Jacob Tsimerman, Victor Wang, and Melanie Matchett Wood, *Remarks on the disproof of the unit distance conjecture*, 2026.
2. T. F. Bloom, *Erdős Problem #1208*, <https://www.erdosproblems.com/1208>, 2026, Accessed 2026-06-14.
3. Marcos Charalambides, *A note on distinct distance subsets*, Journal of Geometry **104** (2013), no. 3, 439–442.
4. Felix Christian Clemen, Jakob Führer, and Oliver Roche-Newton, *Geometric Sidon problems*, 2026.
5. David Conlon, Jacob Fox, William Gasarch, David G. Harris, Douglas Ulrich, and Samuel Zbarsky, *Distinct volume subsets*, SIAM Journal on Discrete Mathematics **29** (2015), no. 1, 472–480.
6. Ernie Croot, Junzhe Mao, Cosmin Pohoata, Adam Sheffer, and Chi Hoi Yip, *A combinatorial large sieve for Sidon sets, distances, and norm forms*, 2026, Preprint.
7. Paul Erdős, *On sets of distances of n points*, American Mathematical Monthly **53** (1946), no. 5, 248–250.
8. Paul Erdős and R. K. Guy, *Distinct distances between lattice points*, Elemente der Mathematik **25** (1970), 121–123.
9. Larry Guth and Nets Hawk Katz, *On the Erdős distinct distances problem in the plane*, Annals of Mathematics **181** (2015), no. 1, 155–190.
10. Farshid Hajir, Christian Maire, and Ravi Ramakrishna, *Cutting towers of number fields*, Annales Mathématiques du Québec **45** (2021), 321–345.
11. Hanno Lefmann and Torsten Thiele, *Point sets with distinct distances*, Combinatorica **15** (1995), no. 3, 379–408.
12. J. S. Milne, *Algebraic number theory*, 2020, Version 3.08, available at <https://www.jmilne.org/math/CourseNotes/ANT.pdf>.
13. Jürgen Neukirch, *Algebraic number theory*, Grundlehren der mathematischen Wissenschaften, vol. 322, Springer-Verlag, Berlin, 1999.
14. OpenAI, *Planar point sets with many unit distances*, 2026, Available from OpenAI.
15. Cosmin Pohoata, *Split primes and the Elekes–Rónyai problem*, 2026.
16. Cosmin Pohoata and Adam Sheffer, *Sets with no isosceles triangles / repeated distances in atlanta*, 2026, Blog post, posted April 27, 2026 and updated June 11, 2026.
17. Luis Ribes and Pavel Zalesskii, *Profinite groups*, 2 ed., Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge, vol. 40, Springer-Verlag, Berlin, 2010.